

Project - Deliverable 1

By: Purabh Singh, Gian Cases, Ryan Amir, Brian Pineda

Schema:

Airline(airline_id:char(2), name:varchar(100) NOT NULL,
primary key (airline_id))

Airport(airport_code:char(3), airport_name:varchar(100) NOT NULL,
city:varchar(100) NOT NULL, country:varchar(100) NOT NULL,
primary key (airport_code))

Operates_From(airline_id:char(2), airport_code:char(3),
primary key (airline_id, airport_code),
foreign key (airline_id) references Airline on delete cascade,
foreign key (airport_code) references Airport on delete cascade)

Aircraft(aircraft_id:int, airline_id:char(2) NOT NULL, model:varchar(50) NOT NULL,
num_economy_seats:int NOT NULL, num_business_seats:int NOT NULL,
num_first_seats:int NOT NULL,
primary key (aircraft_id),
foreign key (airline_id) references Airline on delete cascade)

Flight(airline_id:char(2), flight_number:varchar(10), aircraft_id:int NOT NULL,
departure_airport:char(3) NOT NULL, arrival_airport:char(3) NOT NULL,
departure_time:varchar(10) NOT NULL, arrival_time:varchar(10) NOT NULL,
is_domestic:varchar(5) NOT NULL, operating_days:varchar(20) NOT NULL,
primary key (airline_id, flight_number),
foreign key (airline_id) references Airline on delete cascade,
foreign key (aircraft_id) references Aircraft on delete restrict,
foreign key (departure_airport) references Airport on delete restrict,
foreign key (arrival_airport) references Airport on delete restrict)

Customer(customer_id:int, first_name:varchar(50) NOT NULL,
last_name:varchar(50) NOT NULL, email:varchar(100) NOT NULL,

password:varchar(255) NOT NULL, phone:varchar(20), street:varchar(100),
city:varchar(50), state:varchar(50), zip_code:varchar(10),
primary key (customer_id))

Employee(employee_id:int, first_name:varchar(50) NOT NULL,
last_name:varchar(50) NOT NULL, email:varchar(100) NOT NULL,
password:varchar(255) NOT NULL, phone:varchar(20), role:varchar(20) NOT
NULL,
primary key (employee_id))

Ticket(ticket_number:int, customer_id:int NOT NULL, ticket_type:varchar(20) NOT
NULL, total_fare:float NOT NULL, booking_fee:float NOT NULL,
purchase_datetime:datetime NOT NULL,
primary key (ticket_number),
foreign key (customer_id) references Customer on delete cascade)

Ticket_Flight(ticket_number:int, leg_number:int, airline_id:char(2) NOT NULL,
flight_number:varchar(10) NOT NULL, departure_date:date NOT NULL,
seat_number:varchar(10), class:varchar(20) NOT NULL, special_meal:varchar(50),
primary key (ticket_number, leg_number),
foreign key (ticket_number) references Ticket on delete cascade,
foreign key (airline_id, flight_number) references Flight on delete restrict)

Waiting_List(customer_id:int, airline_id:char(2), flight_number:varchar(10),
flight_date:date, request_datetime:datetime NOT NULL,
primary key (customer_id, airline_id, flight_number, flight_date),
foreign key (customer_id) references Customer on delete cascade,
foreign key (airline_id, flight_number) references Flight on delete cascade)

ER Diagram (Chen Notation):

